

Tech News Tracker – T9.3

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Abstract

The rapid growth of technology journalism across platforms such as TechCrunch, The Verge, and YourStory has made it difficult for users to stay updated efficiently. This project presents an automated tech-news aggregation and summarisation system that collects articles from multiple RSS feeds, classifies them into technology subcategories using a zero-shot Natural Language Inference (NLI) model, and generates concise summaries using a large language model. The system demonstrates how modern NLP pipelines can perform effective classification and summarisation without any task-specific fine-tuning. Experimental evaluation on the India Headlines News Dataset shows that the proposed zero-shot classification pipeline achieves over 90% accuracy on binary tech vs non-tech classification.

1 Introduction

Technology news is produced at an extremely high rate across online media platforms. Readers often spend significant time filtering relevant articles from large streams of information. This project aims to simplify the consumption of technology news by building an intelligent news-tracking application that performs three major tasks:

1. Aggregates live articles from multiple RSS feeds.
2. Classifies articles into meaningful technology subcategories.
3. Generates concise summaries for quick reading.

The main research question explored in this work is:

How effectively can a zero-shot NLI model generalise to technology-news classification without any fine-tuning?

To answer this question, the system is evaluated on a benchmark derived from the India Headlines News Dataset.

2 Dataset

2.1 RSS Feed Sources

The application fetches live news articles from three major technology-news RSS feeds.

Table 1: RSS Feed Sources

Source	Feed URL	Articles/Day
TechCrunch	https://techcrunch.com/feed/	~20
The Verge	https://www.theverge.com/rss/index.xml	~30
YourStory	https://yourstory.com/feed	~15

The system retrieves the 10 most recent articles from each source and sorts them by publication timestamp.

2.2 Evaluation Dataset

For offline evaluation, the India Headlines News Dataset from Kaggle is used. The dataset contains labelled Indian news headlines across multiple categories.

The evaluation benchmark consists of:

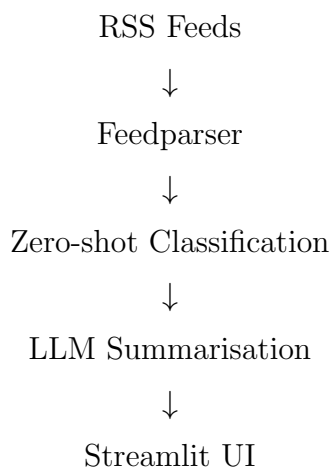
- 150 technology headlines.
- 150 non-technology headlines from entertainment, sports, lifestyle, and world categories.

This produces a balanced binary classification dataset of 300 examples.

3 Methodology

3.1 Pipeline Overview

The overall processing pipeline is shown below:



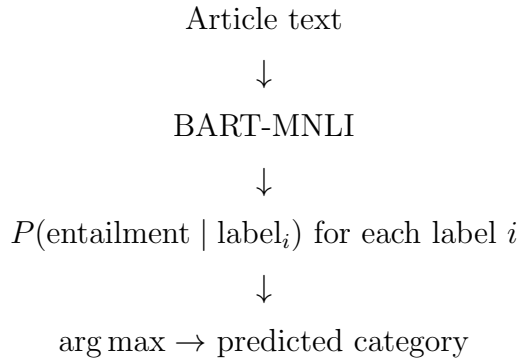
3.2 Zero-Shot Classification

How zero-shot classification works: Traditional text classifiers require labelled training data for every category they must predict. Zero-shot classification removes this requirement by repurposing a Natural Language Inference (NLI) model. NLI models are trained to determine whether a hypothesis sentence is *entailed by*, *contradicted by*, or *neutral to* a premise sentence.

To classify an article into a category, we reformulate the problem as an NLI task:

- **Premise:** “Apple launches new M4 chip for MacBook Pro lineup.”
- **Hypothesis:** “This text is about consumer electronics, gadgets, chips, or hardware devices.”

The model outputs an entailment probability for each candidate hypothesis. The label whose hypothesis receives the highest entailment score becomes the predicted category — with no task-specific training required.



This works because BART-MNLI was fine-tuned on MultiNLI (433k sentence pairs spanning 10 genres), giving it a strong prior over semantic relationships between arbitrary text pairs. The model generalises to unseen label sets at inference time.

We use `facebook/bart-large-mnli` (406M parameters). Given an article text and a set of candidate labels, it estimates the probability that each label is entailed by the text.

Instead of supervised fine-tuning, the model predicts labels using natural-language hypothesis statements. The candidate subcategories are:

Table 2: Technology Subcategories

Display Label	Hypothesis Phrase
AI & Machine Learning	artificial intelligence or machine learning
Startups & Business	technology startups, funding, or business strategy
Gadgets & Hardware	consumer electronics, gadgets, chips, or hardware devices
Software & Apps	software applications, platforms, or developer tools
General Tech	general technology news

The model receives article titles and truncated content as input and produces both predicted labels and confidence scores.

3.3 Summarisation

Article summaries are generated using Llama 3.3 70B through the Groq API. Each summary contains exactly three concise bullet points describing the major contents of the article.

To minimise API calls and reduce latency, generated summaries are cached for one hour.

3.4 Caching Strategy

Table 3: Caching Mechanisms

Component	Mechanism	TTL
Model Weights	@st.cache_resource	Process lifetime
RSS Fetch + Classification	@st.cache_data	15 minutes
Generated Summaries	@st.cache_data	1 hour

4 Results

4.1 Binary Classification Performance

The system was evaluated on the 300-example benchmark.

Table 4: Binary Classification Results

Metric	Value
Accuracy	0.907
Precision	0.906
Recall	0.907
F1 Score	0.906

The results demonstrate that zero-shot NLI models can achieve strong performance on technology-news classification even without fine-tuning.

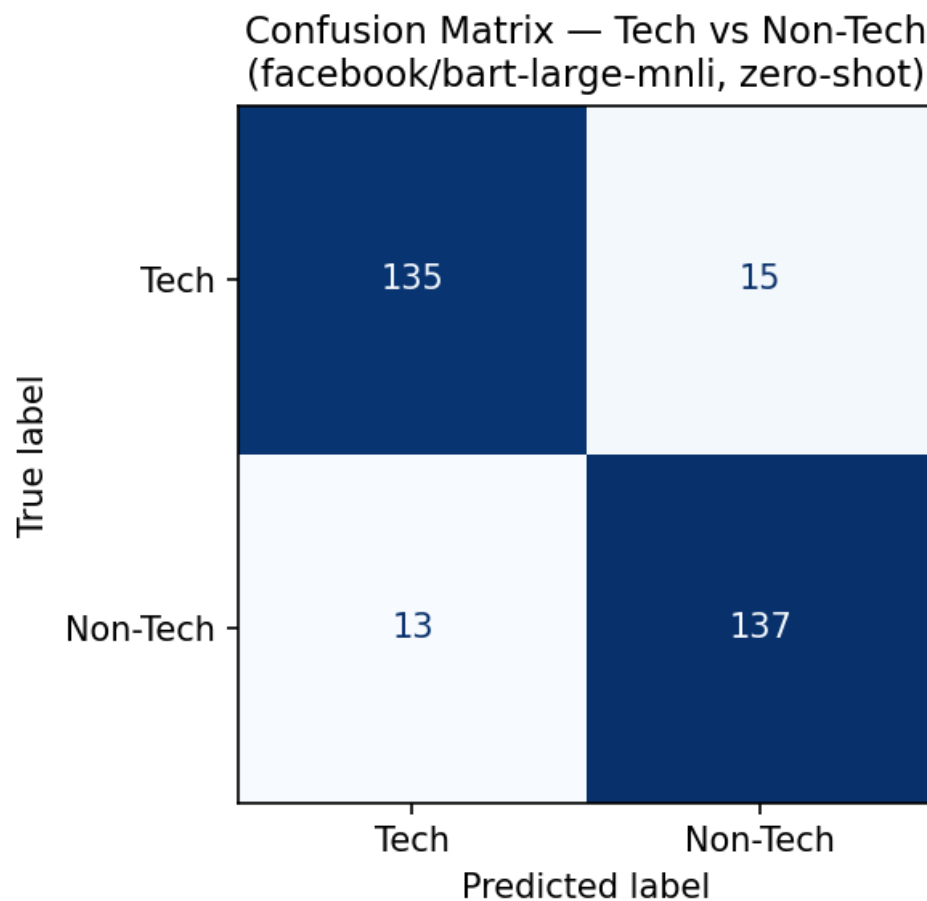


Figure 1: Confusion Matrix (Tech VS Non-Tech)

4.2 Confidence Analysis

Correct predictions are generally associated with high confidence values above 0.85, while incorrect predictions show lower and more scattered confidence values. This behaviour indicates that confidence scores can be useful for uncertainty estimation.

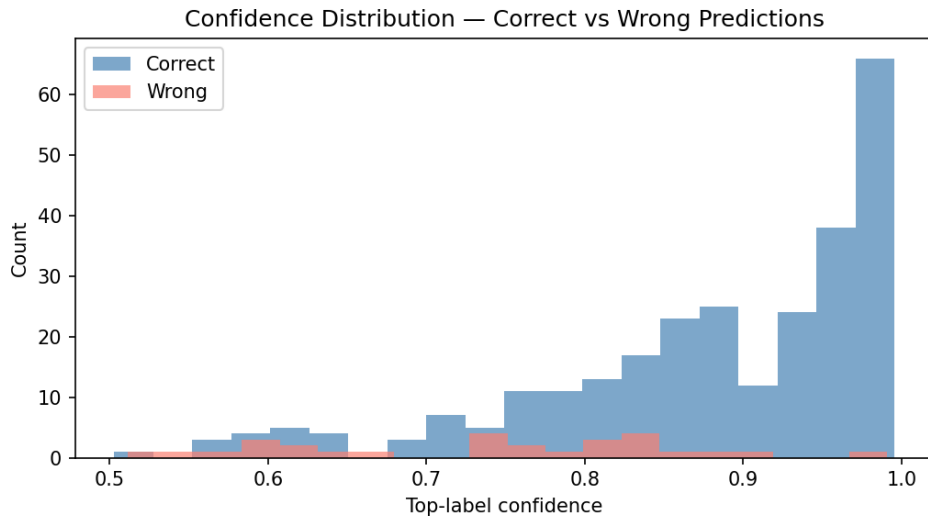


Figure 2: Confidence Distribution

4.3 Subcategory Distribution

The predicted technology articles are distributed across all five technology subcategories. Software-related and general technology news form the majority of the predictions.

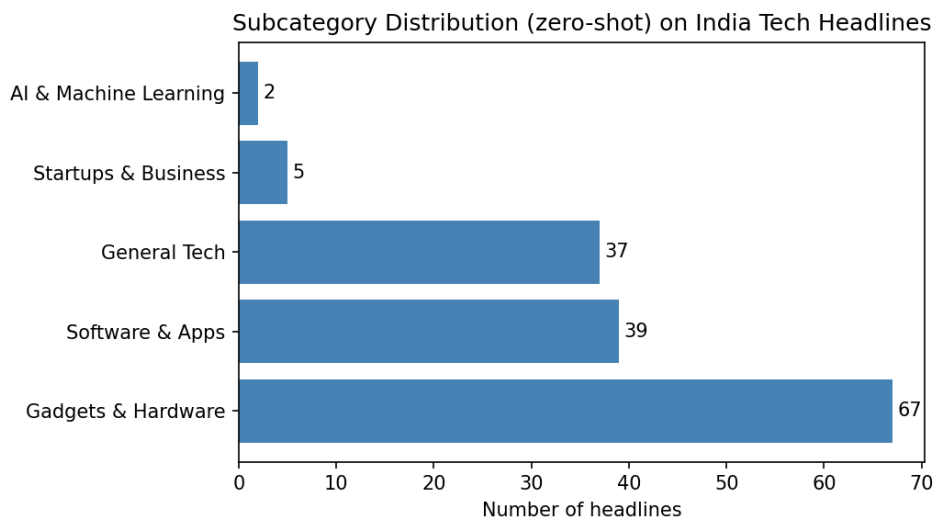


Figure 3: Sub-Category Distribution

5 Ablation Study

An ablation study was conducted to analyse the effect of label phrasing on zero-shot classification performance.

Table 5: Effect of Label Phrasing

Label Style	Example	Accuracy	F1
Descriptive	technology news	0.907	0.906
Short	tech	0.647	0.631
Hypothesis Sentence	This headline is about technology	0.637	0.621

The results show that descriptive labels significantly outperform short keyword-style labels.

6 Limitations

The project has several limitations:

1. The NLI model is trained on general English text and may not fully understand Indian startup terminology or Hinglish.
2. LLM-generated summaries may occasionally hallucinate facts.
3. RSS feeds may become unavailable or change structure unexpectedly.
4. API rate limits restrict large-scale summarisation requests.
5. The model is entirely zero-shot and does not benefit from supervised fine-tuning.

7 Future Work

Future improvements may include:

- Fine-tuning the classifier on technology-news datasets.
- Adding multilingual summarisation support.
- Incorporating recommendation systems based on user preferences.
- Expanding RSS coverage to additional technology publications.

8 Conclusion

This project demonstrates the effectiveness of combining RSS aggregation, zero-shot classification, and large language model summarisation into a unified technology-news tracking platform. Despite requiring no task-specific fine-tuning, the system achieves strong classification performance and provides users with concise, organised technology news updates in real time.

GITHUB link and Live Demo

Live Demo: <https://huggingface.co/spaces/SylvesterSsj/tech-news-terminal>

GitHub: https://github.com/Sylvester4582/Yoraha_Creations_SMAI_A3

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